Intelligent Customer Churn Prediction Using Hybrid Machine Learning and Explainable AI

PROJECT REVIEW 2 — METHODOLOGY, IMPLEMENTATION & RESULTS

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OVERVIEW

What This Review Covers

Implementation evidence, saved experimental results and the current research-paper draft.

01 Methodology

Dataset preparation and end-to-end pipeline

02 System Design

Layered churn analytics architecture

03 Modern Tools & Technologies

Python, machine learning and explainability stack

04 Implementation

Completed modules and verification

05 Results

Baseline and hybrid model outputs

06 Performance Evaluation

Metrics, validation and threshold trade-offs

07 Research Paper Draft

Manuscript status and remaining milestones

METHODOLOGY

Refined Data-to-Insight Pipeline

A reproducible flow from raw customer records to model comparison and risk prediction.

1 **Load & clean**
7,043 rows
21 columns

2 **EDA**
Churn, contract, tenure, charges

3 **Feature engineering**
12 derived features service and charge

4 **Preprocessing**
Imputation, scaling and encoding

5 **Model training**
6 baselines + 6 hybrid pipelines

6 **Evaluation**
5-fold CV + exploratory holdout

7 **Explain & score**
SHAP / importance Dashboard inference

VALIDATION DESIGN

- Stratified 80/20 train–holdout split with random_state = 42.
- Five-fold stratified cross-validation on the training partition.
- Preprocessing and feature engineering are fitted within training folds to reduce leakage.

Caution: the holdout was consulted during candidate development; its scores are exploratory, not an untouched final test.

SYSTEM DESIGN

Repository-Traced Implementation Architecture

The supplied architecture separates offline experimentation, saved artifacts and the interactive Streamlit layer.

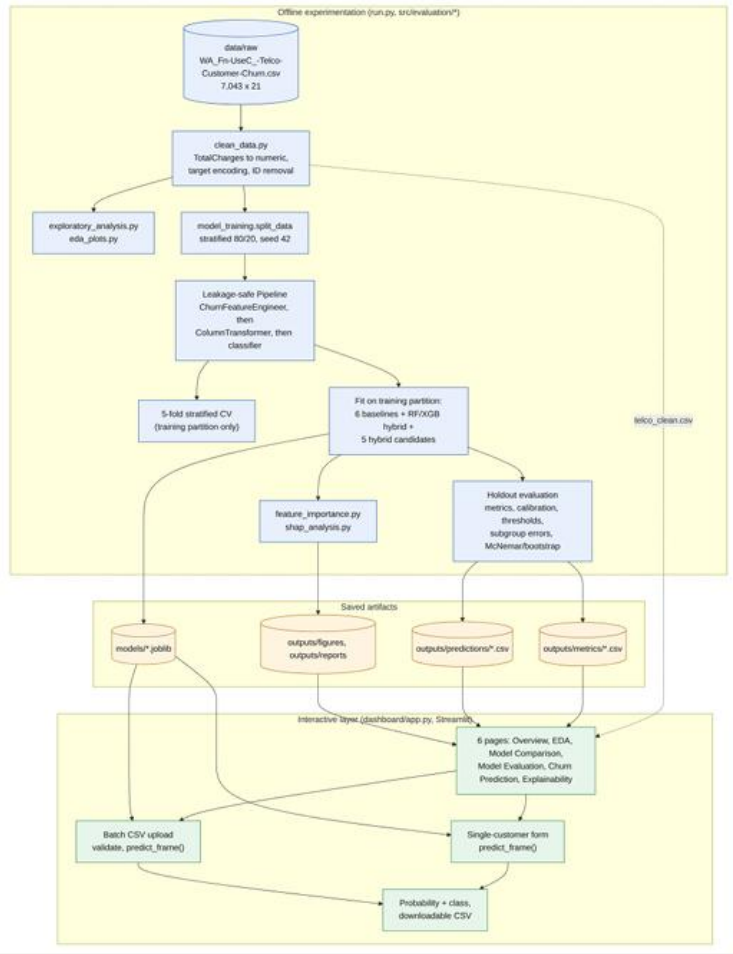


FIG. 1 | BLUE: OFFLINE • ORANGE: ARTIFACTS • GREEN: STREAMLIT

OFFLINE EXPERIMENTATION

The WA_Fn-UseC_-Telco-Customer-Churn.csv input (7,043 × 21) passes through cleaning, EDA, a stratified 80/20 split (seed 42), and a leakage-safe preprocessing pipeline. The training partition supports five-fold stratified CV and fitting of six baselines, the RF–XGBoost hybrid and five additional hybrid candidates. Holdout evaluation records metrics, calibration, thresholds, subgroup errors and paired comparisons.

SAVED ARTIFACTS

Persisted model/joblib files, figures and reports, customer predictions and metrics CSV files form the hand-off between offline experiments and dashboard inference.

INTERACTIVE STREAMLIT LAYER

Six pages: Overview, EDA, Model Comparison, Model Evaluation, Churn Prediction and Explainability. The single-customer form and batch CSV upload both call predict_frame(); the output contains churn probability, predicted class and a downloadable CSV.

Boundary: the dashboard uses the persisted RF–XGBoost scoring model; five extra hybrid candidates and separate calibration / threshold analyses remain offline evaluation outputs.

MODERN TOOLS & TECHNOLOGIES

Technology Stack

Tools are grouped by their role in the implemented workflow.

PY

Python 3.12

Main development and execution environment

DATA

Pandas + NumPy

Tabular data preparation and numerical operations

ML

scikit-learn

Pipelines, preprocessing, classifiers, CV and metrics

XGB

XGBoost

Gradient-boosted tree classifier and hybrid model component

XAI

SHAP + permutation importance

Feature-level interpretation and importance checks

UI

Streamlit + Plotly

Interactive dashboard and prediction visualisation

VIS

Matplotlib + Seaborn

EDA and evaluation charts

OPS

joblib + Git

Persist models and preprocessors; version control

IMPLEMENTATION

Code Implementation and Model Pipeline

The repository contains data preparation, model experiments, evaluation outputs and dashboard modules.

IMPLEMENTED IN REPOSITORY

Dataset	7,043 rows • 21 columns • 1,869 churners (26.54%)
Cleaning	11 blank TotalCharges values converted to numeric; customer ID excluded
Feature engineering	12 derived features: tenure groups, service counts, charge indicators and interactions
Preprocessing	Median/mode imputation, scaling and one-hot encoding within training folds
Models	6 baselines and 6 hybrid/ensemble pipelines
Artifacts	Saved models, metrics, CV outputs, predictions and evaluation figures

VERIFICATION EVIDENCE & BOUNDARIES

- Saved model artifacts loaded successfully during verification.
- All six dashboard pages rendered without exceptions in a scripted session.
- Batch scoring reproduced saved RF+XGBoost probabilities (max absolute difference ≈ 2.2×10^{-16}).
- Required-column validation caught missing input fields; subgroup row-alignment bug was corrected.
- Browser-driven upload/download and formal usability testing were not verified.

RESULTS

Baseline Model Output Snapshot

Selected holdout metrics from the supplied project experiment; higher values are better for both ROC-AUC and PR-AUC.

Model	ROC-AUC	PR-AUC
Logistic Regression	0.8420	0.6370
Random Forest	0.8413	0.6462
XGBoost	0.8428	0.6498

BEST BASELINE ROC-AUC	BEST BASELINE PR-AUC	HOLDOUT SIZE	CHURNERS IN HOLDOUT
0.8428	0.6498	1,409	374

Decision Tree and RBF-SVM trail the leading models on ROC-AUC in this experiment. ANN/MLP has relatively low recall at threshold 0.5 because it is not class-weighted.

RESULTS

Hybrid Candidate Comparison

The highest saved holdout scores are close to the RF–XGBoost baseline and should not be read as proof of a reliable winner.

BEST HOLDOUT ROC-AUC	BEST HOLDOUT PR-AUC	RF + XGBOOST PR-AUC	RF MEAN CV ROC-AUC
0.8443	0.6573	0.6510	0.8459

WEIGHTED RF / XGBOOST / SVM VOTE

ROC-AUC 0.8443

PR-AUC 0.6573

RF + XGBOOST HYBRID

ROC-AUC 0.8439

PR-AUC 0.6510 | Recall 0.7647 at threshold 0.5

Paired tests with multiplicity adjustment do not establish statistically significant hybrid superiority. Differences are small relative to fold-level variability.

PERFORMANCE EVALUATION

How Models Are Compared

A multi-dimensional evaluation avoids choosing a model from accuracy alone.

01 CLASSIFICATION

Accuracy • Precision • Recall
Specificity • F1

02 RANKING

ROC-AUC • PR-AUC

03 AGREEMENT

MCC • Cohen's Kappa

04 PROBABILITY QUALITY

Log loss • Brier score
Expected Calibration Error

05 STABILITY

5-fold stratified CV
Mean and standard deviation

06 DECISION TRADE-OFF

Threshold analysis • False negatives
Calibration and retention cost

Thresholds should be selected on validation data using explicit retention costs, not optimized against the already-inspected holdout.

PERFORMANCE EVALUATION

Validation, Thresholds and Statistical Caution

The validation design is intended to reduce leakage and expose trade-offs, not to overstate small score differences.

VALIDATION PROTOCOL

- Stratified 80/20 split with random_state = 42.
- Five-fold stratified cross-validation on training data only.
- Preprocessing and feature engineering fitted inside each fold.
- Paired model comparisons use exact McNemar tests and paired bootstrap intervals for PR-AUC differences.

THRESHOLD TRADE-OFF

RF + XGBoost

Threshold	Recall	False positives
0.50	0.765	244
0.30	0.888	406

Higher recall catches more churners but flags more non-churners for retention outreach.

The holdout was consulted during candidate development; confirmatory evaluation requires a fresh split or nested CV.

RESEARCH PAPER DRAFT PREPARATION

Manuscript Status and Next Milestones

The paper consolidates the experimental design, model comparison, validation checks and repository audit.

CURRENT MANUSCRIPT

Comparative Evaluation of Hybrid Models for Telecom Customer Churn Prediction: Discrimination, Calibration, and Error Analysis

Draft sections

Introduction • Related Work • Methodology • Experimental Setup • Results • Discussion • Limitations • Conclusion and Future Work

Draft prepared: 25 October 2026 | Not submitted

REMAINING MILESTONES

- Complete training-only tuning.
- Reconcile all figures and result tables.
- Verify SHAP outputs and save artifacts.
- Run a fresh/temporal confirmatory evaluation.
- Finalize dataset citation and reference metadata.
- Confirm author details and obtain supervisor review.

Source register is compiled; all references and dataset metadata require a final verification pass before submission.

IMPLEMENTATION REVIEW

Challenges Addressed and Remaining Work

The review distinguishes verified behavior from work that still needs a controlled final check.

Class imbalance	Used precision–recall metrics and threshold analysis, not accuracy alone.
Potential leakage	Fitted learned preprocessing inside training-fold pipelines.
Subgroup mismatch	Corrected row alignment using customer ID and regenerated subgroup analysis.
Calibration	Added Brier score, log loss, ECE and reliability-curve analysis.
Reused holdout	Clearly labelled holdout metrics exploratory; fresh evaluation recommended.
Reproducibility	Documented environment versions, saved artifacts and stale-output checks.

Future work: additional datasets, temporal validation, cost-sensitive thresholds, subgroup fairness review and dashboard usability study.

Thank You

Questions & Feedback

Intelligent Customer Churn Prediction

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